

In our setting, let $L/\mathbb{Q}$ be a Galois extension, i.e. L is the splitting field of some polynomial in $\mathbb{Q}[x]$.

More generally, everything works in the following setting:

- R is a Dedekind domain with finite residue fields,
- $K = \text{Frac}(R)$,
- L/K is a finite extension,
- S is the integral closure of R in L .

We study how a prime ideal $\mathfrak{p}$ of R factors in S :

$$\mathfrak{p}S = \mathfrak{p}_1^{e_1} \cdots \mathfrak{p}_r^{e_r},$$

where $\mathfrak{p}_i$ are distinct primes of S .

Recall .0.1

We already know that if p is a prime number, then

$$p\mathcal{O}_L = \mathfrak{p}_1^{e_1} \cdots \mathfrak{p}_r^{e_r}$$

where $\mathfrak{p}_i$ are distinct prime ideals of $\mathcal{O}_L$ and e_i are positive integers. Moreover, let

$$f_i := [\mathcal{O}_L/\mathfrak{p}_i : \mathbb{F}_p].$$

Then:

$$[L : K] = \sum_{i=1}^r e_i f_i.$$

Now fix L/K as Galois and let

$$G := \text{Gal}(L/K).$$

Remark .0.1

Every $\sigma \in G$ maps $\mathcal{O}_L \rightarrow \mathcal{O}_L$ bijectively.

Proof. Let $\alpha \in \mathcal{O}_L$, then α is integral over $\mathbb{Z}$. Thus it satisfies a monic polynomial with coefficients in $\mathbb{Z}$. Applying σ , we see $\sigma(\alpha)$ satisfies the same polynomial, hence is integral. Thus $\sigma(\mathcal{O}_L) \subseteq \mathcal{O}_L$. Applying the same argument to σ^{-1} gives equality. Since σ is a field automorphism, it is bijective. $\mathbb{Z}$

Remark .0.2

This statement tells us that σ permutes the primes $\mathfrak{p}_i$ in the factorization of $p\mathcal{O}_L$.

Lemma .0.1

G acts transitively on the set $\{\mathfrak{p}_1, \dots, \mathfrak{p}_r\}$.

Proof. Fix $\mathfrak{p}_i$. Suppose $\sigma(\mathfrak{p}_i) \neq \mathfrak{p}_j$ for all $\sigma \in G$. By the Chinese Remainder Theorem, there exists $x \in \mathcal{O}_L$ such that

$$\begin{aligned} x &\equiv 0 \pmod{\mathfrak{p}_j} \\ x &\equiv 1 \pmod{\sigma(\mathfrak{p}_i)} \quad \forall \sigma \in G. \end{aligned}$$

Consider the norm:

$$N_{L/\mathbb{Q}}(x) = \prod_{\sigma \in G} \sigma(x) \in \mathbb{Z} \cap \mathfrak{p}_j = p\mathbb{Z}.$$

(Note here we use the assumption that L/K is Galois). But $\sigma(x) \notin \mathfrak{p}_i$ for all σ , so the product is not in $\mathfrak{p}_i$. But $p\mathbb{Z} \subset \mathfrak{p}_i$, a contradiction. $\mathbb{Z}$

Corollary .0.2

All e_i are equal and all f_i are equal:

$$e_1 = \cdots = e_r = e, \quad f_1 = \cdots = f_r = f.$$

Proof. Since G acts transitively, for any i, j there exists σ such that

$$\sigma(\mathfrak{p}_i) = \mathfrak{p}_j.$$

Applying σ to the factorization:

$$p\mathcal{O}_L = \prod \mathfrak{p}_i^{e_i}$$

gives

$$p\mathcal{O}_L = \prod \sigma(\mathfrak{p}_i)^{e_i}.$$

Thus exponents must match, so $e_i = e_j$.

Also, σ induces an isomorphism:

$$\mathcal{O}_L/\mathfrak{p}_i \cong \mathcal{O}_L/\mathfrak{p}_j,$$

through the map

$$\mathcal{O}_L \xrightarrow{\sigma} \mathcal{O}_L \xrightarrow{\pi} \mathcal{O}_L/\mathfrak{p}_j.$$

then the kernel is $\mathfrak{p}_i$, so $\mathcal{O}_L/\mathfrak{p}_i \cong \mathcal{O}_L/\mathfrak{p}_j$. This gives $f_i = f_j$. $\mathbb{Z}$

Corollary .0.3

When L/G is Galois, we conclude that

$$p\mathcal{O}_L = (\mathfrak{p}_1 \cdots \mathfrak{p}_r)^e$$

where $f_i = [\mathcal{O}_L/\mathfrak{p}_i : \mathbb{F}_p] = f$ (does not depend on $\mathfrak{p}_i$). So

$$[L : \mathbb{Q}] = efr$$

Definition .0.1

The **decomposition group** of $\mathfrak{p}_i$ is the stablizer of $\mathfrak{p}_i$ in G . i.e.

$$D_i := \{\sigma \in G \mid \sigma(\mathfrak{p}_i) = \mathfrak{p}_i\}.$$

Remark .0.3

By orbit-stabilizer:

$$r = [G : D_i].$$

For $\sigma \in D_i$, we also get an isomorphism

$$\mathcal{O}_L/\mathfrak{p}_i \cong \mathcal{O}_L/\mathfrak{p}_i$$

Fact:

If ℓ is a finite field, then $\#\ell = p^k$ for some $k \in \mathbb{Z}_{\geq 0}$, and p a prime number, and

$$\text{Aut}(\ell) = \{\sigma_i \mid i \in \mathbb{Z}/k\mathbb{Z}\}$$

there $\sigma_i : x \rightarrow x^{p^i}$. (This is called the **Frobenius automorphism**.)

In our setting, $L/\mathbb{Q}$ is Galois with group G . If D is the decomposition group of some $\mathfrak{p}_i$ of $\mathcal{O}_L$ which contains p , then $\sigma \in D$ induces $\bar{\sigma}$

$$\bar{\sigma} : \text{Aut}(\mathcal{O}_L/\mathfrak{p}_i)$$

This yields a homomorphism:

$$D \rightarrow \text{Aut}(\mathcal{O}_L/\mathfrak{p}_i)$$

by

$$\bar{\sigma}(x + \mathfrak{p}_i) = \sigma(x) + \mathfrak{p}_i$$

Proposition .0.4

The homomorphism $\theta : D \rightarrow \text{Aut}(\mathcal{O}_L/\mathfrak{p}_i)$ as above is surjective.

Proof. Proved later.

$\mathbb{Z}$

Definition .0.2

The **inertia group** is

$$I := \ker(\theta).$$

Corollary .0.5

From the isomorphism, we get

$$|D| = |I| \cdot |\text{Aut}(\mathcal{O}_L/\mathfrak{p}_i)|$$

We know that $|\text{Aut}(\mathcal{O}_L/\mathfrak{p}_i)| = f$ from the previous fact, and $|D| = ef$. We therefore conclude that $|I| = e$.

Remark .0.4

- D/I is cyclic group of order f .
- $f = [D : I]$
- $r = [G : D]$
- $e = |I|$, and I is a cyclic group of order e .